IPL 2025 Dashboard Project

Solution Report: Insights, Method and Recommendations

Source: IPL_Dashboard_Project.xlsx (60 matches, 22 Mar to 19 May 2025, 8 teams, 10 venues, 33 players, 201 player-match rows).

1. Executive summary

• **SRH and CSK lead on results**: SRH won 9 of 13 decided matches (69.2%), CSK 8 of 12 (66.7%). PBKS (33.3%), DC (35.7%) and MI (38.5%) are at the bottom.

• **Batting-heavy RCB underdelivers**: RCB has the best average score (182.6) and plays at the highest-scoring ground (Chinnaswamy, 185.7), yet wins only 54.5%. It also concedes the most runs (180.6 per match).

• **The home side won 60%** of decided matches (30 of 50). In this dataset the home side (Team1) also bats first, so the two effects cannot be separated.

• **The toss did not decide matches**: toss winners won only 30% (15 of 50). Winning the toss was no advantage in this season.

• **Virat Kohli is the standout**: 982 runs at an average of 65.5 and strike rate 162.6, about 1.6 times the next batter. He also leads Man of the Match awards (4).

• **Finishers hit hardest**: Andre Russell (SR 189.7) and Heinrich Klaasen (SR 188.3) score 71 to 75% of their runs in boundaries.

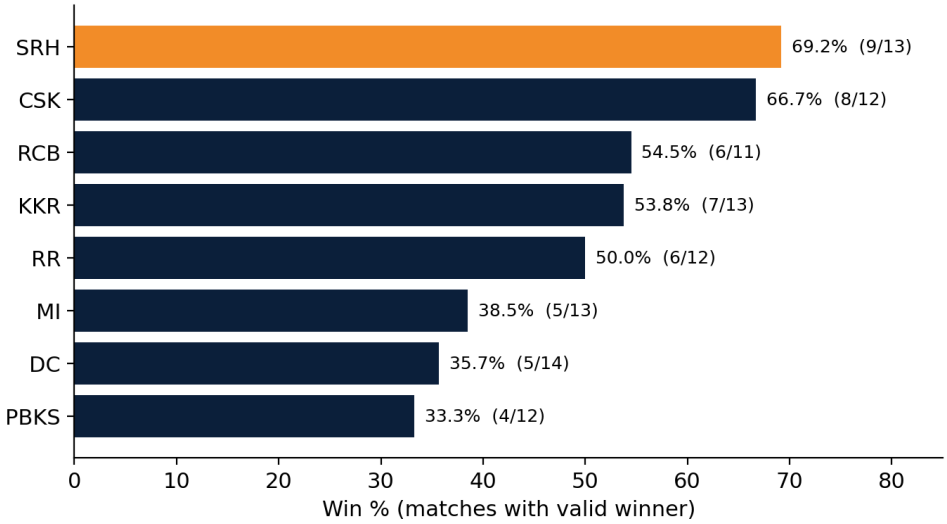
• **Bumrah leads bowling** with 25 wickets at economy 6.95; Kuldeep Yadav is the tightest regular bowler (economy 4.60).

• **Data issues found**: 10 of 60 matches list a winner who did not play; the dashboard strike-rate KPI is understated (see section 6).

2. Questions answered

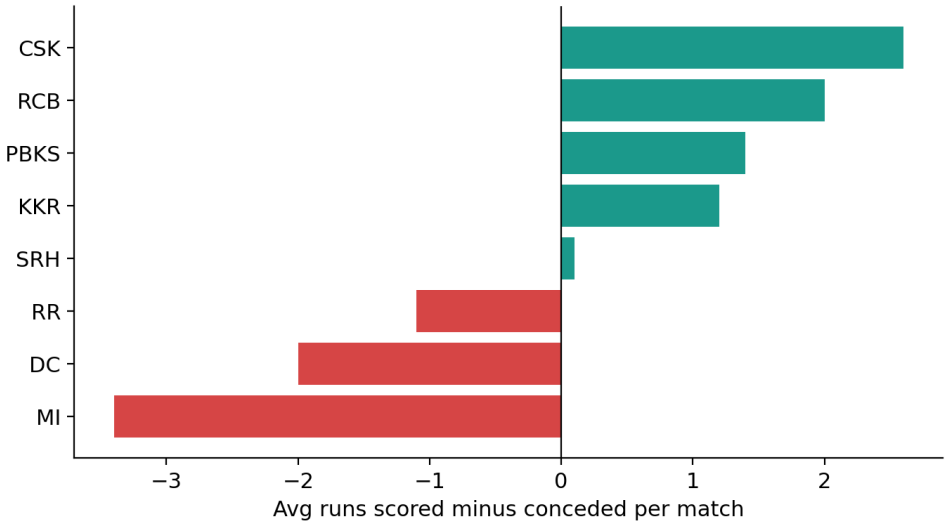
Question	Answer
Which team won most?	SRH by win % (9/13, 69.2%); CSK next (8/12)
Who scored most runs (team)?	RR 2,755 total; RCB best per match (182.6)
Best batter?	Virat Kohli: 982 runs, avg 65.5
Best strike rate (3+ inn.)?	Andre Russell 189.7; Klaasen 188.3
Top wicket-taker?	Jasprit Bumrah: 25 wkts
Best economy (10+ overs)?	Kuldeep Yadav: 4.60
Highest-scoring venue?	Chinnaswamy, Bengaluru: 185.7 avg
Lowest-scoring venue?	Sawai Mansingh, Jaipur: 165.6 avg
Does toss matter?	No: toss winners won 30% (15/50)
Does home advantage exist?	Home side won 60% (30/50)
Most Man of the Match?	Kohli (4); eight players tied at 3
Average innings score?	174.5 runs; only 3 scores of 200+ (max 208)

3. Team performance



Win % uses the 50 matches whose recorded winner is one of the two teams playing.

Team	Matches	Decided	Wins	Win %	Avg scored	Avg conceded	Net/match
SRH	15	13	9	69.2	171.9	171.8	0.1
CSK	14	12	8	66.7	176.6	174.0	2.6
RCB	15	11	6	54.5	182.6	180.6	2.0
KKR	15	13	7	53.8	176.9	175.7	1.2
RR	16	12	6	50.0	172.2	173.3	-1.1
MI	16	13	5	38.5	170.8	174.2	-3.4
DC	16	14	5	35.7	171.4	173.4	-2.0
PBKS	13	12	4	33.3	174.7	173.3	1.4

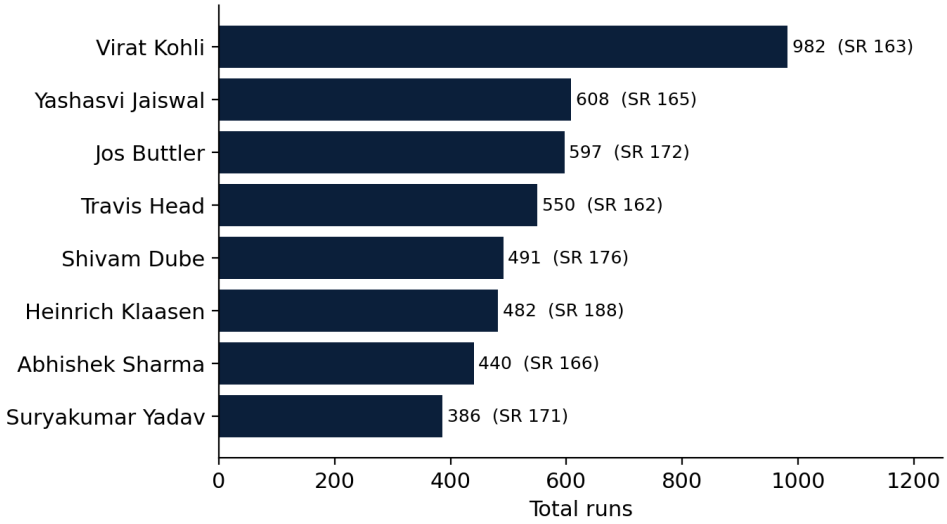


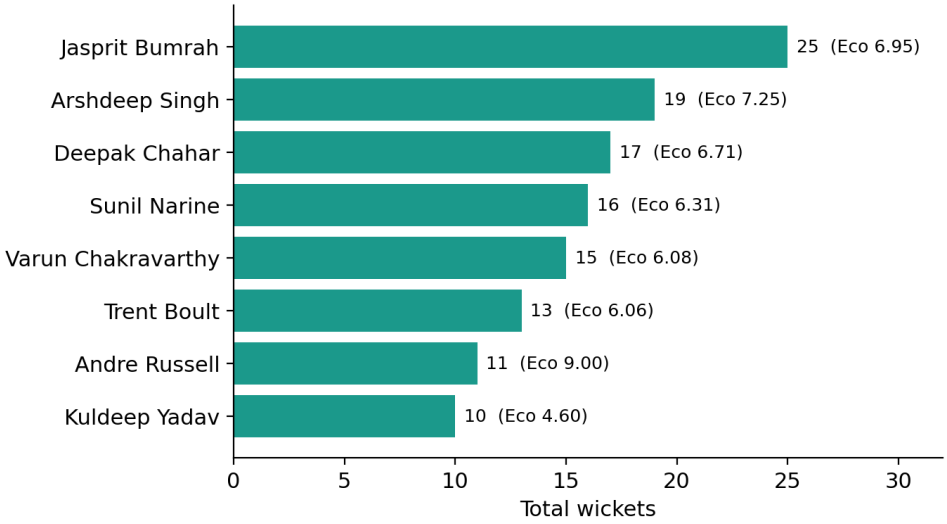
• CSK (+2.6) and RCB (+2.0) have the best run margin per match; MI (-3.4) and DC (-2.0) the worst.

• SRH wins 69% with a margin near zero (+0.1), a sign of close-match wins that may be hard to repeat.

• PBKS has a positive margin (+1.4) but the lowest win rate: results are not following scoring.

4. Player performance

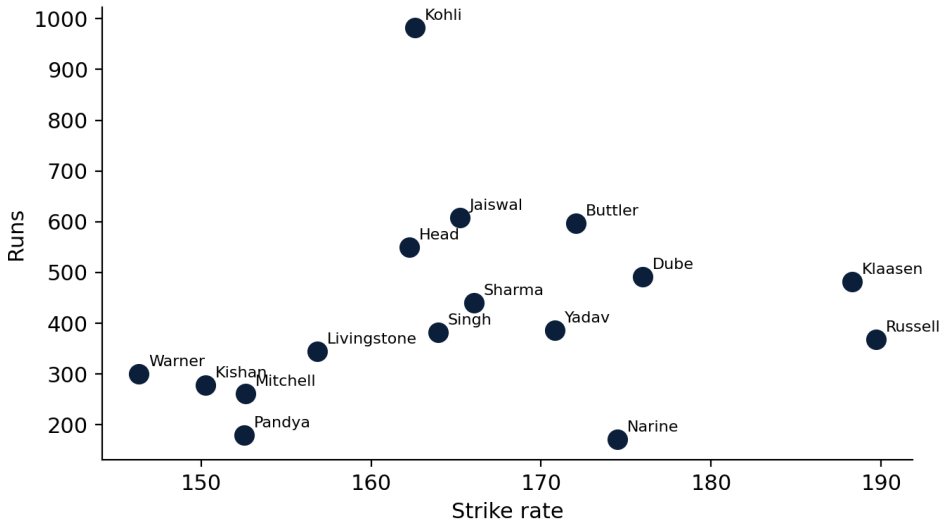




Batter	Runs	Balls	Inn.	4s	6s	SR
Virat Kohli	982	604	15	94	34	162.6
Yashasvi Jaiswal	608	368	9	59	21	165.2
Jos Buttler	597	347	8	59	22	172.0
Travis Head	550	339	8	58	21	162.2
Shivam Dube	491	279	8	49	23	176.0
Heinrich Klaasen	482	256	7	41	30	188.3

Bowler	Wkts	Runs conc.	Overs	Economy
Jasprit Bumrah	25	267	38.4	6.95
Arshdeep Singh	19	232	32.0	7.25
Deepak Chahar	17	188	28.0	6.71
Sunil Narine	16	202	32.0	6.31
Varun Chakravarthy	15	113	18.6	6.08
Trent Boult	13	114	18.8	6.06

Strike rate vs volume



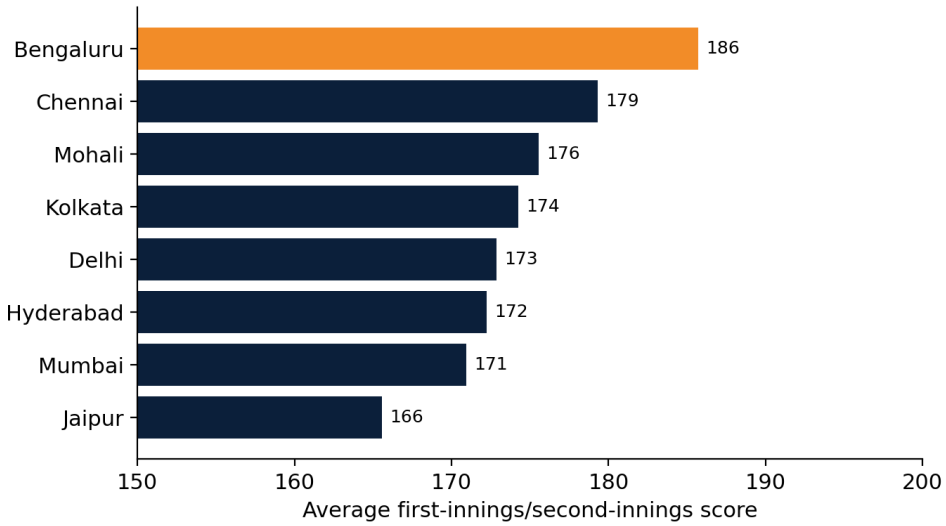
• Batting roles: wicketkeepers have the highest strike rate (171), then all-rounders (168) and batsmen (163).

• Bowlers take 132 of the 178 recorded wickets (74%).

• Six-hitting: 320 sixes and 690 fours; sixes provide 25.7% of all runs in the player data.

• Wicket-taking quality: Varun Chakravarthy has the best bowling average (7.5 runs per wicket) among leading bowlers; Russell is expensive (economy 9.00).

5. Venues



• Bengaluru (185.7) and Chennai (179.3) favour batters; Jaipur (165.6) and Mumbai (170.9) are lowest.

• Delhi and Mohali were won by the batting-first side in only 2 of 5 and 2 of 7 decided matches. Samples are small (5 to 7), so treat as indicative.

6. Data quality and limitations

• **Invalid winners**: 10 of 60 matches have a Match_Winner who was not one of the two teams (for example M047, M048, M051, M053, M055, M057, M058). They were excluded from win %, toss and home analysis. Run totals in 7 matches also disagree with the recorded winner. Please verify source data.

• **Strike-rate KPI is understated**: the workbook averages Strike_Rate over all 201 rows, including bowling-only rows with 0. That gives 106.7. The correct figure (total runs / total balls) is 166.0.

• **Economy KPI**: the dashboard sums the Economy column (613.4), which is not meaningful. Use total runs conceded / total overs = 7.08.

• **Wickets KPI**: team-level wickets (799) are wickets lost by batting sides summed across innings, whereas player-level wickets total 178. Player rows cover only part of each match (about 3 players per match), so player leaderboards are a sample, not the full season.

• **Small samples**: 13 to 16 matches per team and 5 to 8 per venue; conclusions are indicative.

7. Recommendations

• **RCB**: invest in bowling; the batting is already the best in the league.

• **MI, DC, PBKS**: results lag; review death-over bowling and match-up planning.

• **Team strategy**: do not rely on the toss; plan for ground conditions, and use venue averages to set targets (about 185 at Bengaluru, 165 at Jaipur).

• **Auction/selection**: prioritise high-strike-rate finishers (Russell, Klaasen) and economical spinners (Kuldeep, Varun).

• **Dashboard fixes**: use weighted strike rate and economy, filter bowling-only rows from batting averages, and add a data check for the winner field.

8. Method

Python (pandas) joined fact_match, fact_player, dim_team, dim_player and dim_venue. Win % = wins / decided matches. Average scored/conceded = runs per team innings. Strike rate = runs / balls x 100. Economy = runs conceded / overs. Batting tables use rows with balls faced above 0; bowling tables rows with overs above 0. Full steps are in README.md.